

Homework (Carolina Reaper)

- Q1.** A painter mixes x gallons of blue paint and y gallons of yellow paint to create a shade for a canvas of size $3x \cdot y$. If $x^2 + y^2 = 25$ and $x + y = 5$, what is the size of the canvas?
(A) 20
(B) 25
(C) 30
(D) 35
(E) 40
- Q2.** For a music rhythm pattern, the number of beats x and the number of rests y are related by $x^2 - 2y^2 = 12$ and $2x + 3y = 14$. What is the value of x ?
(A) 2
(B) 3
(C) 4
(D) 5
(E) 6
- Q3.** A sound wave has a frequency x and amplitude y . If $x^2 + y^2 = 36$ and $x - y = 2$, what is the amplitude of the sound wave?
(A) 2
(B) 3
(C) 4
(D) 5
(E) 6
- Q4.** An artist creates a sculpture of size $x \cdot y$, where x and y satisfy $x^2 + 2y^2 = 48$ and $2x - y = 8$. What is the size of the sculpture?
(A) 24
(B) 32
(C) 40
(D) 48
(E) 56
- Q5.** A DJ mixes two songs, one with a frequency x and the other with a frequency y . If $x^2 - y^2 = 16$ and $x + y = 8$, what is the frequency of the mixed song?
(A) 4
(B) 5
(C) 6
(D) 7
(E) 8
- Q6.** A guitar string vibrates at a frequency x and has a length y . If $x^2 + y^2 = 49$ and $x - 2y = -3$, what is the length of the guitar string?
(A) 2
(B) 3
(C) 4
(D) 5
(E) 6
- Q7.** A painter creates a mosaic of size $x \cdot y$, where x and y satisfy $x^2 - 3y^2 = 24$ and $2x + 3y = 18$. What is the size of the mosaic?
(A) 24
(B) 30
(C) 36
(D) 42
(E) 48
- Q8.** A drum has a resonant frequency x and a diameter y . If $x^2 + 2y^2 = 72$ and $x + 2y = 12$, what is the resonant frequency of the drum?
(A) 4
(B) 5
(C) 6
(D) 7
(E) 8

Open Ended Questions (FRQ)

- Q9.** Solve the system of equations $x^2 - 2y^2 = 15$ and $2x + 3y = 17$. Show all work and explain your solution.
- Q10.** Graph the system of equations $x^2 + y^2 = 25$ and $x - 2y = -3$. Identify the points of intersection and explain their significance in the context of a musical composition.

Chef's Tips

- Tip 1: Make sure to check for extraneous solutions.
- Tip 2: Use the quadratic formula when necessary.
- Tip 3: Encourage students to verify their solutions graphically.